WEEK 10: REAL-ESTATE ANALYTICS DASHBOARD

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1. Introduction

The Real-Estate Analytics Dashboard is developed using Microsoft Power BI to analyze house sales data from King County. The dashboard helps real-estate managers understand property prices, geographical locations, house characteristics, and sales patterns.

Interactive visualizations, DAX measures, geographical maps, and numeric range parameters are used to provide meaningful insights into the real-estate market.

2. Problem Statement

Real-estate managers need an effective way to understand property prices, locations, house characteristics, and sales patterns. An interactive Power BI dashboard is developed to analyze these factors and support better decision-making.

3. Objectives

- To calculate total properties sold, average house price, median price, average living area, and average price per square foot.
- To create geographical visualizations using latitude and longitude.
- To analyze the impact of property grade, condition, and waterfront availability on house prices.
- To implement numeric range parameters for dynamic price and square-footage filtering.

- To develop an interactive House Sales Analytics Dashboard.

4. Dataset Description

Dataset Name: King County House Sales Dataset

The dataset contains 21,613 property records and 21 columns.

Important attributes include:

- Price – Selling price of the property.
- Bedrooms – Number of bedrooms.
- Bathrooms – Number of bathrooms.
- Sqft Living – Interior living area.
- Waterfront – Indicates waterfront availability.
- Grade – Construction and design quality.
- Condition – Overall condition of the property.
- Latitude and Longitude – Geographical location of properties.
- Zipcode – Property location code.

The dataset was preprocessed during Week 9 before being used for dashboard development.

5. Tools and Technologies Used

- Microsoft Power BI Desktop
- Power Query Editor
- DAX (Data Analysis Expressions)
- Azure Maps
- King County House Sales Dataset

6. Data Preparation

The dataset was cleaned and transformed during Week 9 using Power Query Editor.

The following transformations were performed:

- Checked and corrected data types.
- Created Sale Year and Sale Month columns.
- Calculated Property Age.
- Created Price per SqFt.
- Created Price Category.
- Created Renovation Status.
- Created Condition Category.
- Created Living Area Category.
- Created Waterfront Status.
- Calculated Total Area.
- Created Property Size Bucket.

The prepared dataset was reused for the Week 10 dashboard.

7. DAX Measures

The following DAX measures were created to calculate important real-estate metrics.

1. Total Properties Sold

Total Properties Sold =

COUNTROWS('Housing_Data')

2. Average House Price

Avg House Price =

AVERAGE('Housing_Data'[price])

3. Median Price

Median Price =

MEDIAN('Housing_Data'[price])

4. Average Living Area

Avg Living Area =

AVERAGE('Housing_Data'[sqft_living])

5. Average Price per SqFt

Avg Price per SqFt =

AVERAGE('Housing_Data'[Price per SqFt])

These measures were used to create KPI cards in the dashboard.

8. Numeric Range Parameters

Two numeric range parameters were created to support interactive filtering.

Living Area Filter

- Minimum: 0
- Maximum: 14,000
- Increment: 100
- Default: 2,000

Price Filter

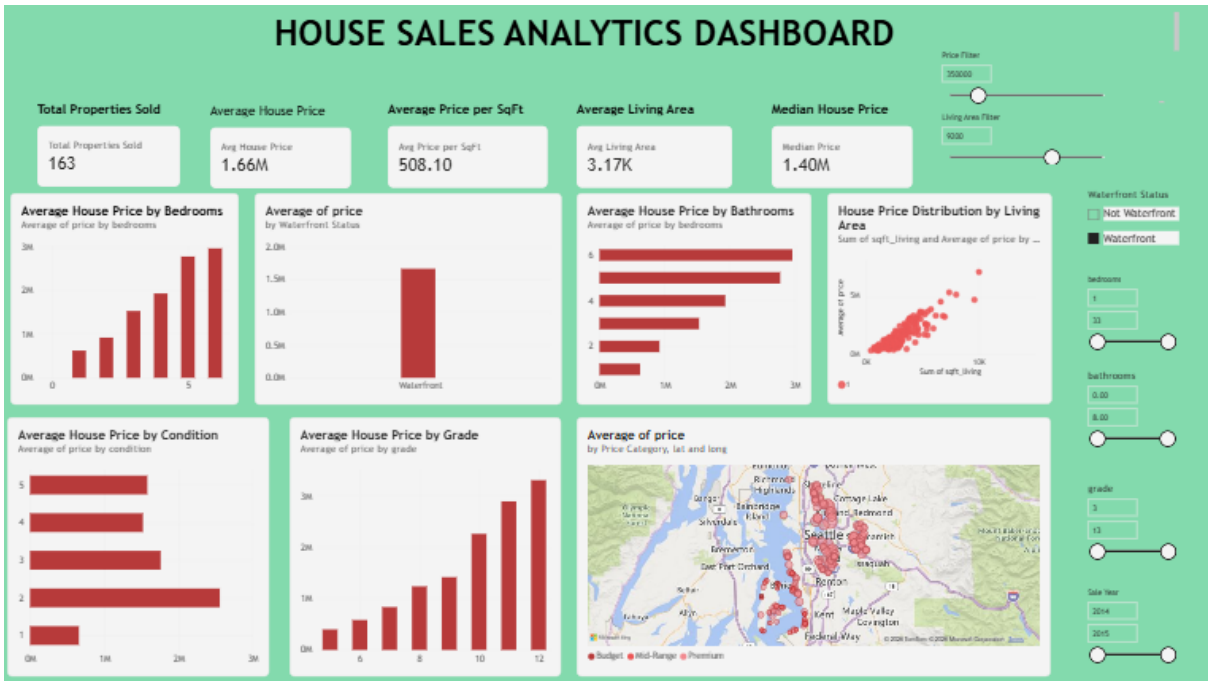
- Minimum: 0
- Maximum: 2,000,000

- Increment: 50,000
- Default: 500,000

These parameters allow users to adjust the living area and property price thresholds dynamically.

9. Dashboard Visualizations

The following visualizations were developed:



1. KPI Cards

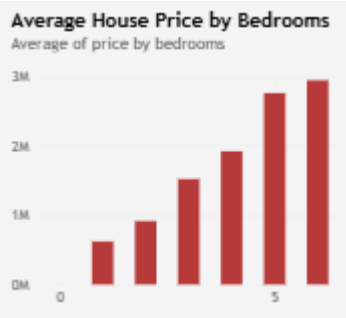
Five KPI cards display:

- Total Properties Sold
- Average House Price
- Median House Price
- Average Living Area
- Average Price per SqFt



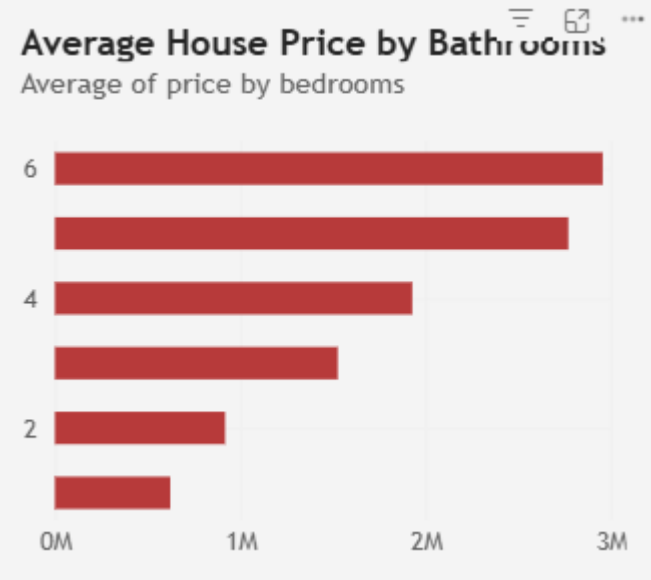
2. Average House Price by Bedrooms

A column chart is used to compare average house prices based on the number of bedrooms.



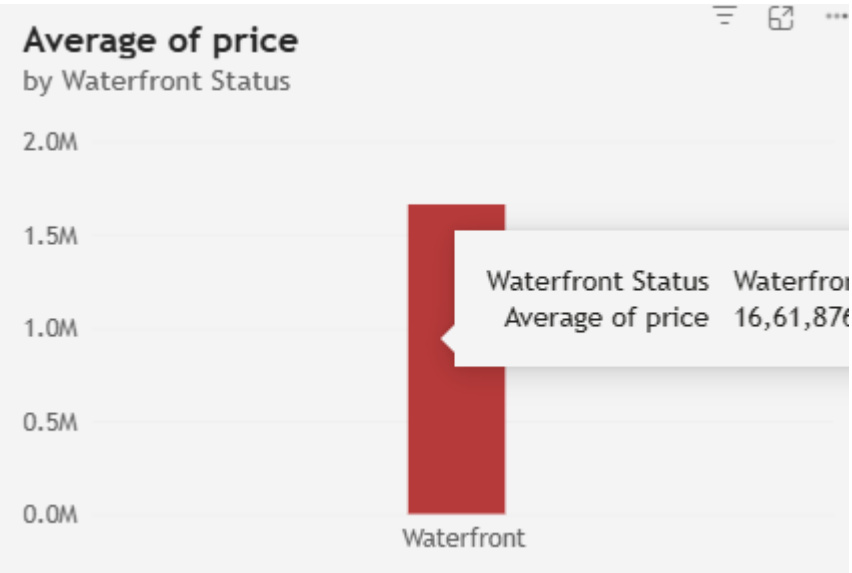
3. Average House Price by Bathrooms

A bar chart is used to analyze the relationship between bathroom count and average property price.



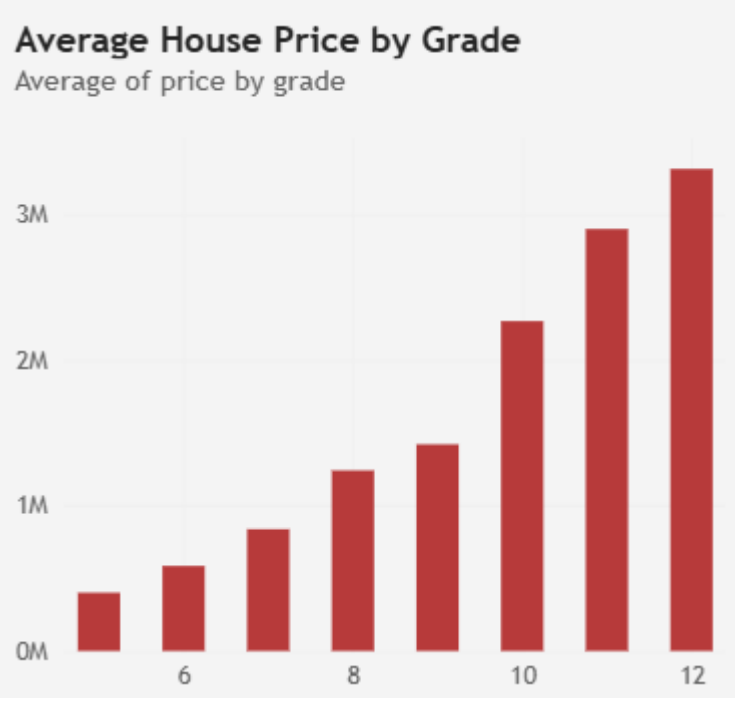
4. Waterfront Comparison

A column chart compares the average prices of waterfront and non-waterfront properties.



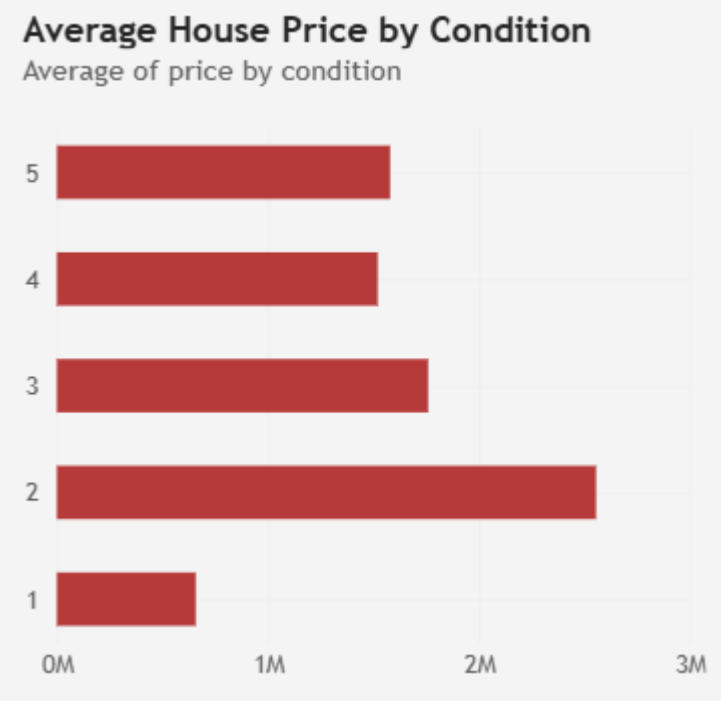
5. House Price by Grade

A column chart analyzes the impact of property grade on average house prices.



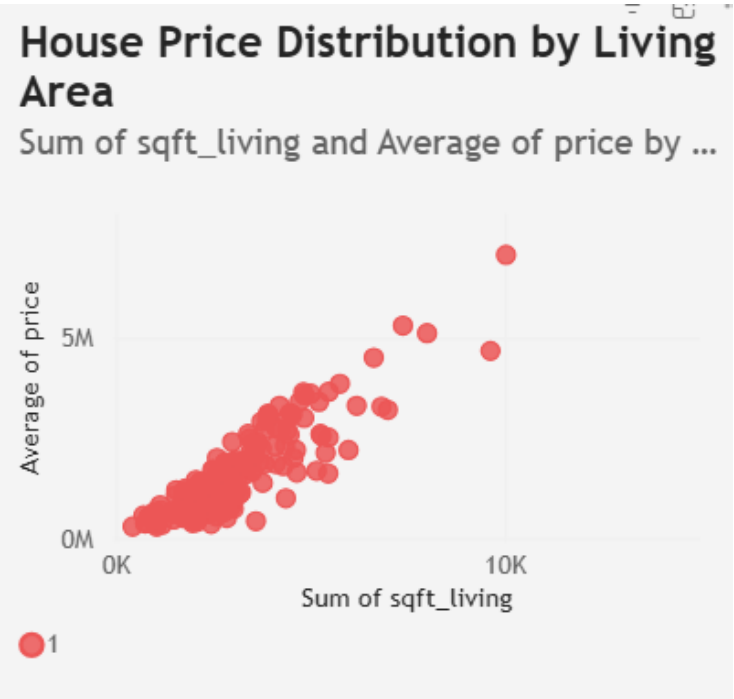
6. House Price by Condition

A bar chart compares property prices across different condition categories.



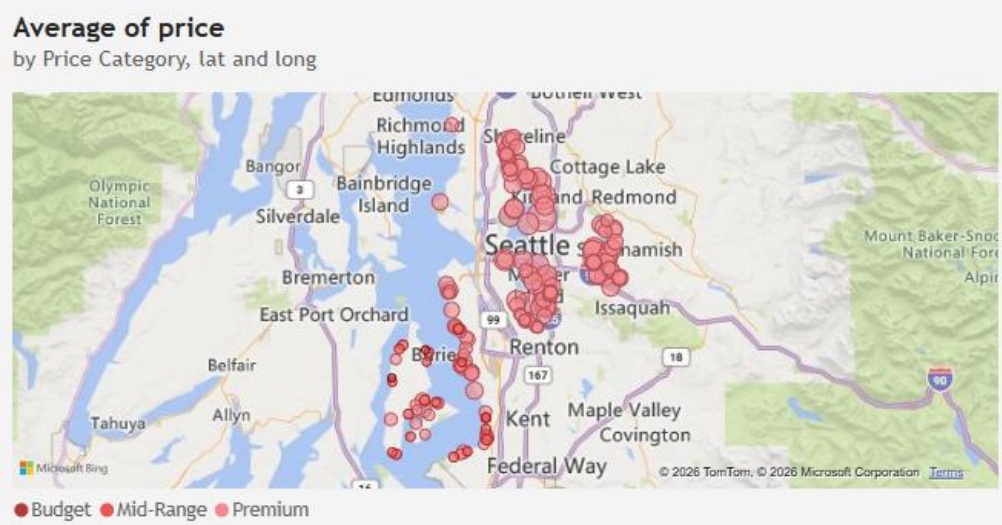
7. Price Distribution by Living Area

A scatter chart visualizes the relationship between living area and house prices.



8. Geographical Map

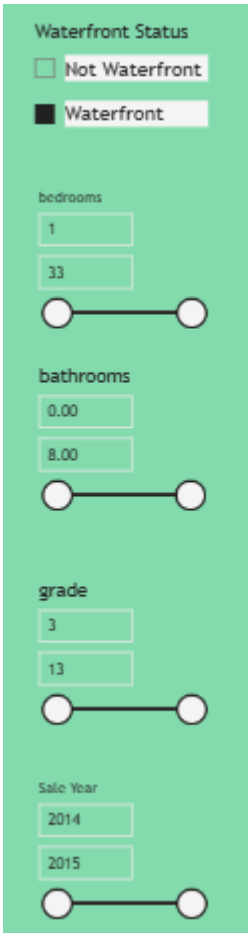
Azure Maps is used to visualize property locations using latitude and longitude. Different price categories are represented using different colours.



10. Interactive Slicers

The following slicers were added to improve dashboard interactivity:

- Waterfront Status
- Bedrooms
- Bathrooms
- Property Grade
- Sale Year



The numeric range parameter sliders allow users to explore properties based on selected living area and price thresholds.

11. Results and Analysis

The dashboard provides an interactive view of the King County real-estate market.

- KPI cards provide a quick summary of property sales and price statistics.
- Bedroom and bathroom charts help compare property prices based on house characteristics.
- Waterfront analysis helps identify price differences between waterfront and non-waterfront properties.

- Grade and condition charts show how property quality relates to selling prices.
- The geographical map displays the spatial distribution of properties.
- Scatter chart analysis helps understand the relationship between living area and house prices.
- Numeric parameters provide flexible filtering based on price and living area.

12. Conclusion

The Real-Estate Analytics Dashboard was successfully developed using Microsoft Power BI. DAX measures, interactive slicers, numeric range parameters, charts, and geographical mapping were used to analyze the King County House Sales Dataset.

The dashboard provides useful insights into property prices, geographical distribution, house characteristics, and sales patterns. This project demonstrates how Power BI can transform raw real-estate data into meaningful and interactive visual information for better decision-making.